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SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0329 – MOBILE COMPUTING

PROJECT TITLE

"Wireless Ad Hoc Network Visualizer & Emergency Messaging App"

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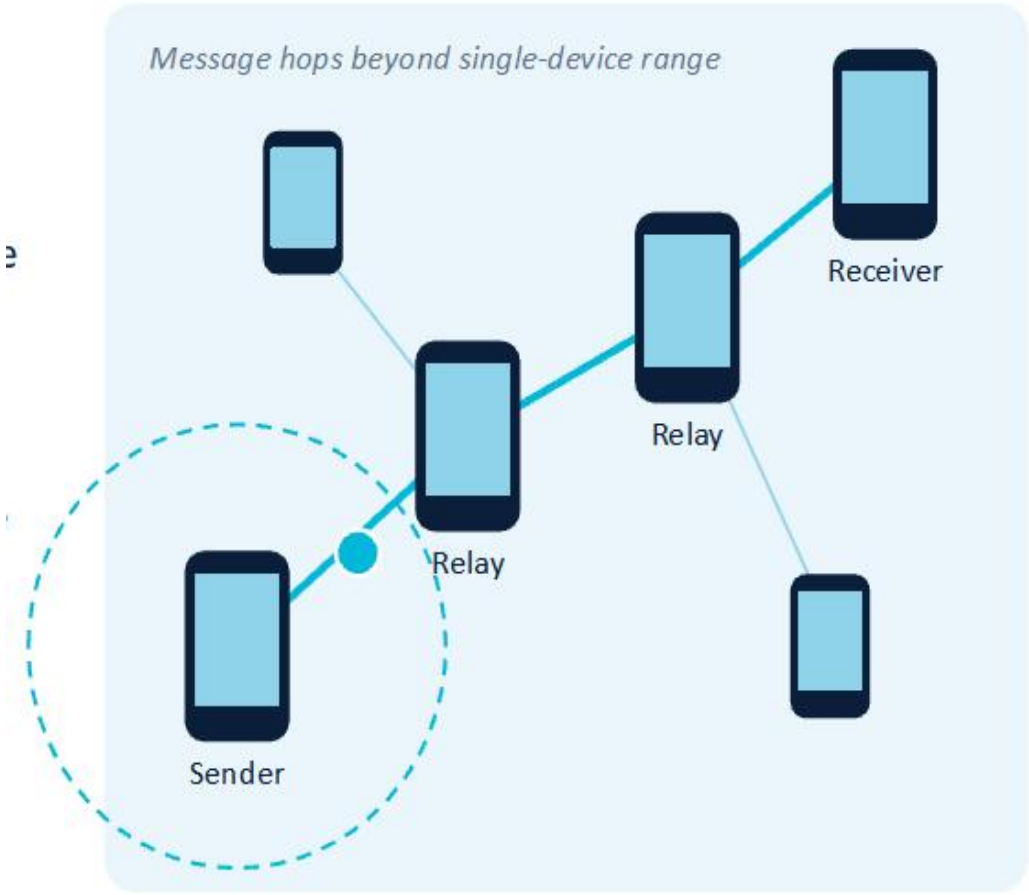
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ABSTRACT

- This project presents an Android-based wireless ad hoc network visualizer and emergency messaging application for nearby devices.
- Nearby devices are discovered and connected using the Wi-Fi P2P and Nearby Connections APIs, and the network is shown as a graph of nodes and links.
- Emergency messages are forwarded across relay devices by an application-layer routing component that uses TTL and message IDs to limit propagation and duplicates.
- Undelivered messages are stored locally in a Room database and retried when connectivity returns, if the app is permitted to run.
- Planned evaluation measures are message delivery rate, hop count, message delay, network lifetime, and battery consumption.

INTRODUCTION

- A wireless ad hoc network connects devices directly, with no central access point
- Devices communicate through nearby peers
- Multi-hop routing carries messages beyond the range of a single connection
- Emergency messaging supports communication during outages and disasters
- The Android app visualizes connectivity and message delivery



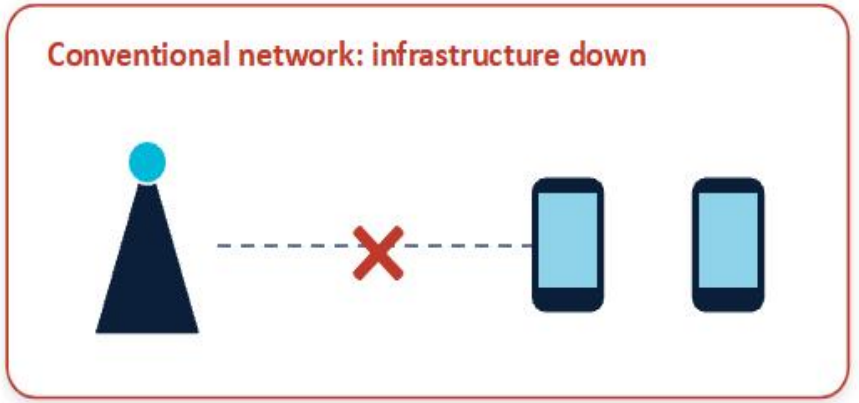
PROBLEM STATEMENTS AND OBJECTIVES

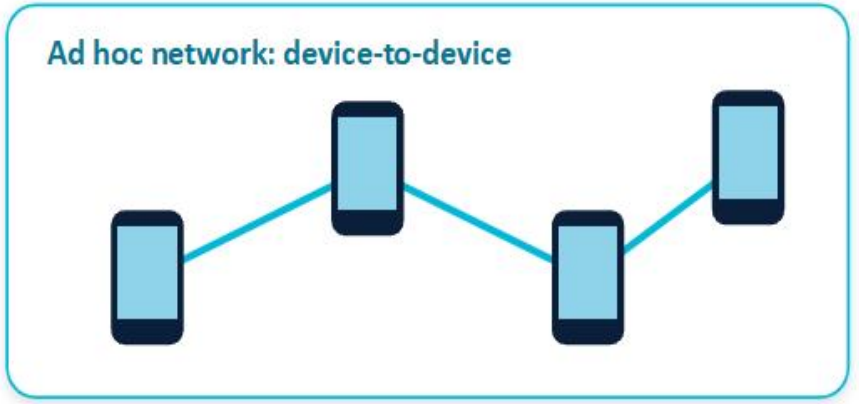
Problem Statement:

- Conventional messaging depends on available infrastructure
- Network outages can interrupt emergency communication
- Limited wireless range restricts direct device-to-device messaging

Objectives:

- Discover nearby devices using supported peer-to-peer APIs.
- Establish direct wireless connections between devices
- Implement multi-hop message forwarding using TTL
- Implement multi-hop message forwarding usinEvaluate message delivery rate, hop count, network lifetime, and battery consumptiong TTL





EXISTING SYSTEM AND PROPOSED SYSTEMS

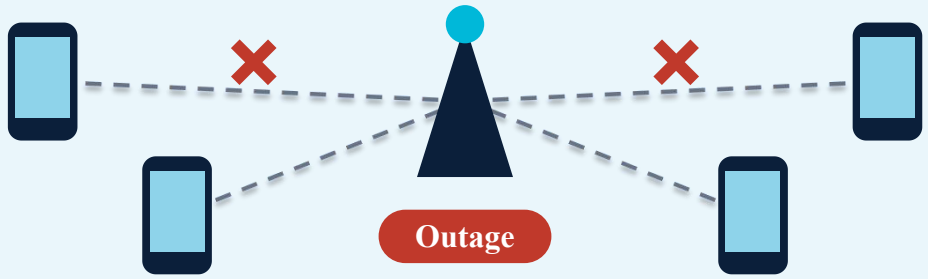
Existing System

- Conventional messaging depends on cellular or internet infrastructure.
- Communication can fail during network outages and disasters.
- Direct device-to-device communication has limited range.
- Device movement can interrupt communication paths.

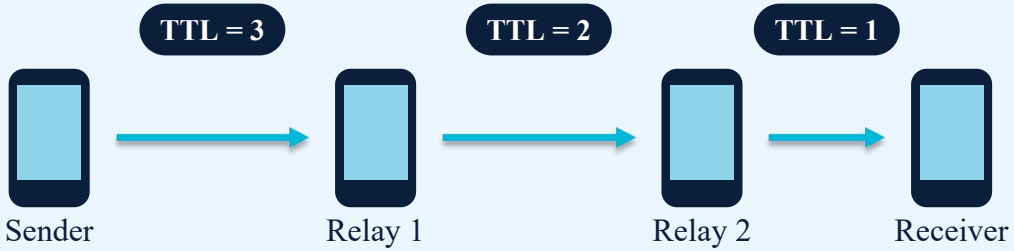
Proposed System

- Android-based peer-to-peer communication using supported wireless APIs.
- Multi-hop emergency message forwarding through connected devices.
- Time-to-Live (TTL) limits unnecessary message propagation.
- Store-and-forward delivery using local message storage.

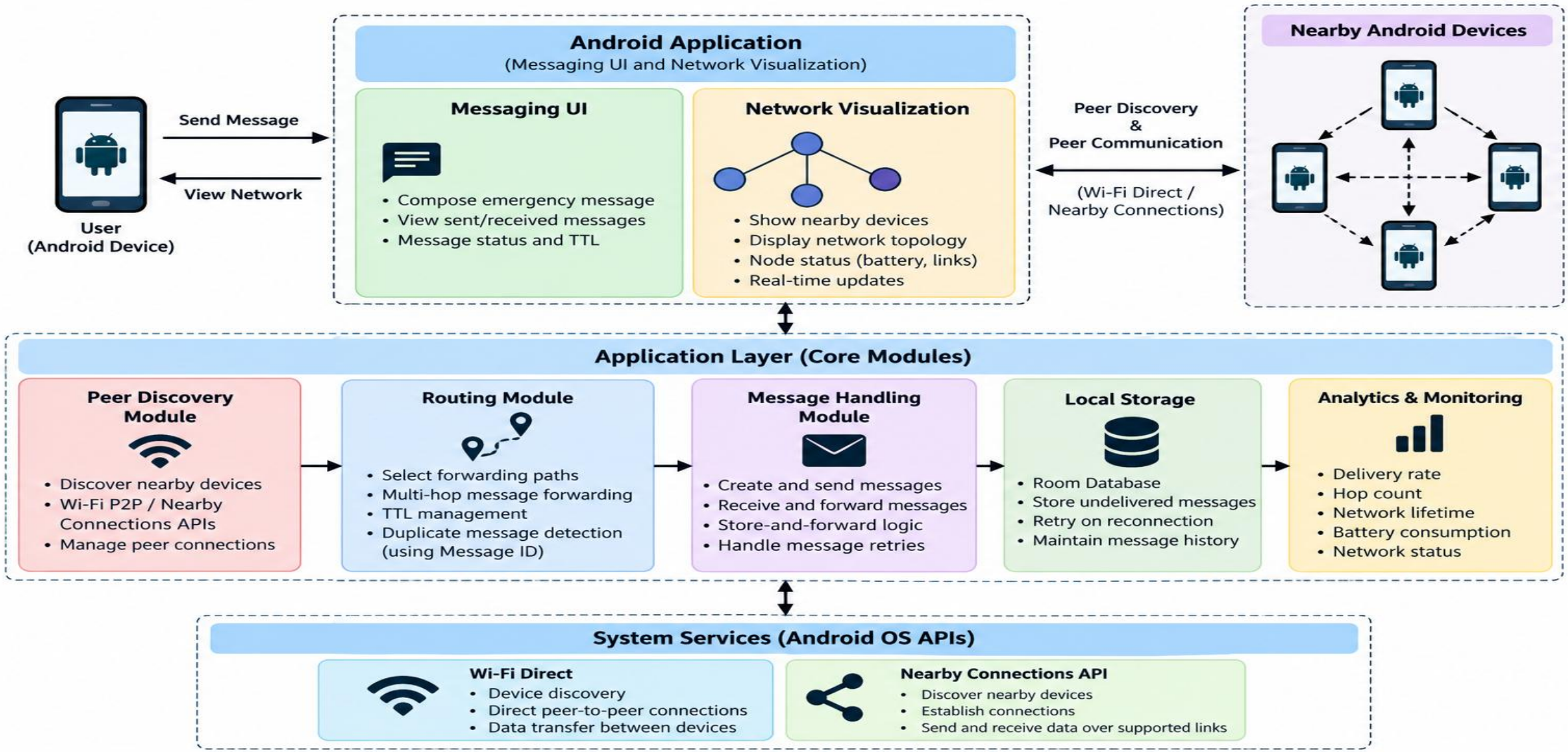
Conventional: infrastructure-dependent



Proposed: device-to-device multi-hop

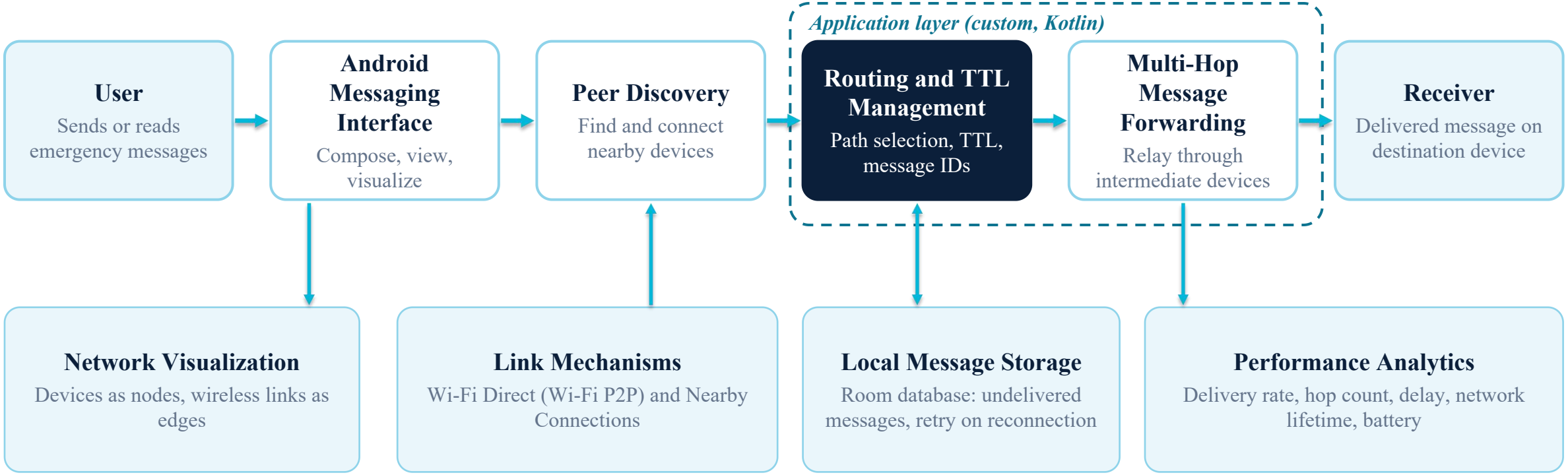


METHODOLOGY



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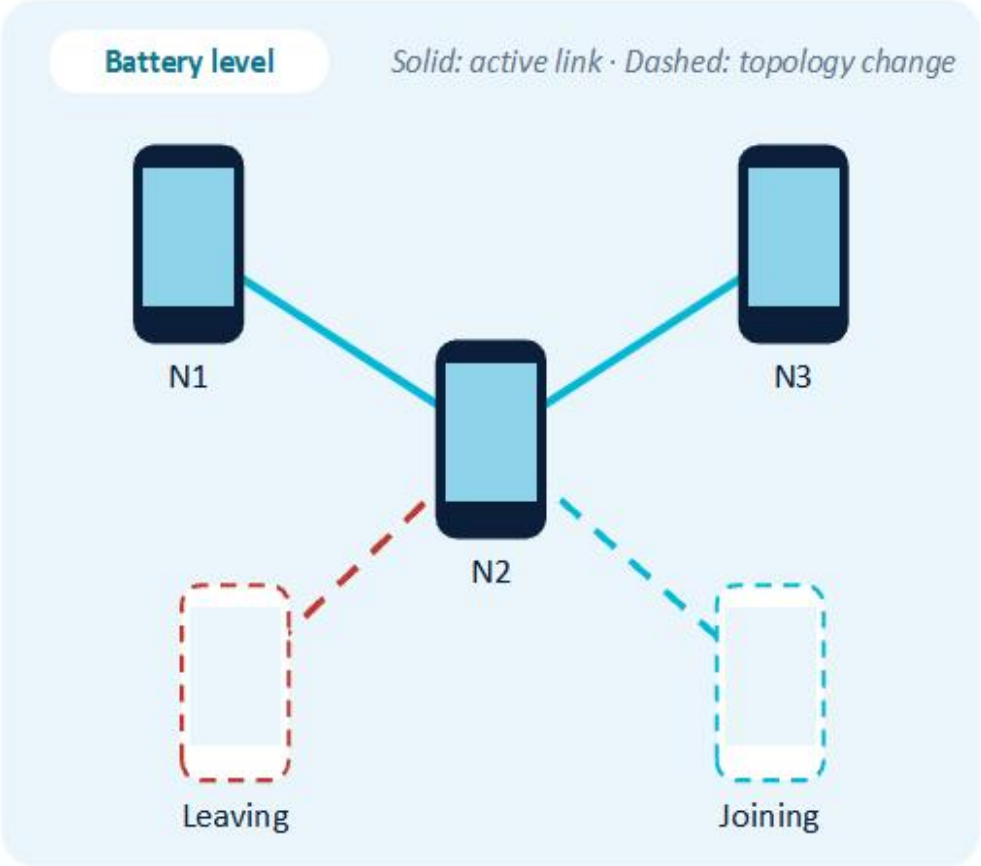
METHODOLOGY – WORKFLOW



Wi-Fi Direct and Nearby Connections are alternative or complementary link mechanisms. Multi-hop routing is implemented at the application layer; Android does not automatically route messages across arbitrary peers.

Module -1: Peer Discovery & Visualization

- Discover nearby devices via Wi-Fi P2P or Nearby Connections
- Establish and monitor supported peer connections
- Nodes represent devices; edges represent wireless links
- Update the graph when devices connect or disconnect
- Display configurable node count, connection range, battery level, and topology changes



Module 2: Multi-Hop Emergency Messaging



- Create emergency messages and send to connected peers
- Relay through a custom routing layer; TTL limits propagation
- Message IDs enable duplicate detection and stop repeat forwarding
- Undelivered messages are stored in Room and retried on reconnection, if the app is permitted to run
- Evaluate delivery rate, hop count, delay, network lifetime, and battery use

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RESULTS

Message Delivery Rate

Messages delivered ÷ messages sent

Measured value: to be recorded

Hop Count

Number of relay hops per delivered message

Measured value: to be recorded

Message Delay

Time from send to receipt

Measured value: to be recorded

Network Lifetime

Time until a set share of nodes become unavailable or connectivity is lost

Measured value: to be recorded

Battery Consumption

Battery used by a node during a test run

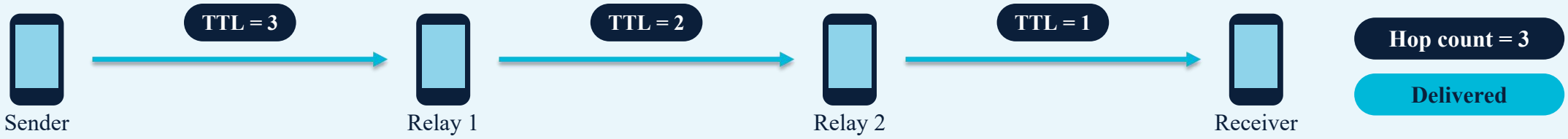
Measured value: to be recorded

Network Status

Connected nodes and active links

Measured value: to be recorded

ILLUSTRATIVE EXAMPLE – NOT MEASURED DATA



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IEEE STANDARDS

IEEE STANDARDS

IEEE 802.11-2024
Wireless LAN MAC and PHY specifications; the standard behind Wi-Fi-based peer links.

IEEE 802.15.1-2005
Historical Bluetooth-based WPAN standard (inactive–withdrawn). Background only; not implemented here.

IETF RFCs (not IEEE)

RFC 3561 – AODV
Ad hoc On-Demand Distance Vector routing (Experimental). Reference for possible future routing enhancements.

RFC 4728 – DSR
Dynamic Source Routing for IPv4 (Experimental). Reference for possible future routing enhancements.

IEEE standards define wireless link technologies; IETF RFCs define routing protocols. This project uses a custom application-layer flooding-with-TTL router and does not implement AODV or DSR.

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SDG CONTRIBUTION

SDG 9
Industry, Innovation and Infrastructure

Demonstrates an alternative communication approach for situations where conventional infrastructure is unavailable.

SDG 11
Sustainable Cities and Communities

Supports emergency communication and disaster resilience in communities.

SDG 13
Climate Action

Indirect, potential contribution only: if applied to climate-related disasters, it could support communication during such events.

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CONCLUSION

- Enables emergency communication between supported nearby Android devices without relying on conventional network infrastructure.
- Combines peer discovery, multi-hop routing, message storage, and network monitoring to support resilient emergency messaging.



REFERENCES

[1] IEEE Std 802.11-2024, IEEE Standard for Information Technology, Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications. IEEE, 2025.

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